

STUDENT NUMBER

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MATHEMATICS EXTENSION ONE TRIAL HIGHER SCHOOL CERTIFICATE

WEDNESDAY 18th JULY 2012

General Instructions

- Reading Time – 5 minutes
- Working time – 2 hours
- Write using black or blue pen
- Board-approved calculators may be used
- A table of standard integrals is provided at the back of this paper
- All necessary working should be shown in every question
- Start a new booklet for each question

Total Marks - 75

- Attempt questions 1 - 14
- Answer questions 1 – 10 on the multiple choice answer sheet provided
- For questions 11-14, start each question in a new booklet

QUESTION NO	MARK
1 - 10	/10
11	/15
12	/15
13	/15
14	/15
TOTAL	/70

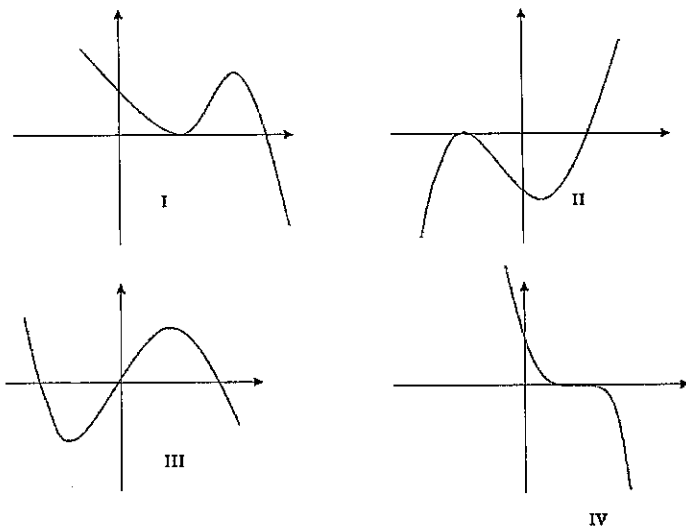
THIS QUESTION PAPER MUST NOT BE REMOVED FROM THE EXAMINATION ROOM
This assessment task constitutes 40% of the Higher School Certificate Course Assessment.

Answer questions 1 – 10 on the multiple choice answer sheet provided 1 MARK EACH

- The coordinates of the point P that divides the interval joining (-1,3) and (4,8) internally in the ratio 3:2 is
 - (6,2)
 - (2,6)
 - (1,5)
 - (5,1)
- The value of $\lim_{x \rightarrow 0} \frac{\sin 3x}{2x}$
 - $\frac{2}{3}$
 - $\frac{1}{2}$
 - $\frac{1}{6}$
 - $\frac{3}{2}$
- The Cartesian equation of the curves whose parametric equations are $x = \sin t$ and $y = \cos^2 t + 1$ is
 - $y = x^2 - 1$
 - $y = 1 - x^2$
 - $y = 2 - x^2$
 - $y = x^2 - 2$
- The values of x for which $\frac{x^2 - 9}{x} < 0$ is
 - $-3 < x < 3$
 - $x < -3$ and $0 < x < 3$
 - $x < -3$ and $x > 3$
 - $-3 < x < 0$ and $x > 3$

5. A polynomial equation $f(x) = 0$ has a double root.

Which of the following are possible graphs of $y = f(x)$?



- (A) I or II
(B) II or III
(C) III or IV
(D) II or III or IV

6. A function $f(x)$ is given by the equation $f(x) = 2 \sin^{-1}(3x)$.

The domain and the range of $f(x)$ are respectively:

- (A) $-3 \leq x \leq 3$ and $-2 \leq f(x) \leq 2$.
(B) $-\frac{1}{3} \leq x \leq \frac{1}{3}$ and $-\frac{\pi}{2} \leq f(x) \leq \frac{\pi}{2}$
(C) $-\frac{1}{3} \leq x \leq \frac{1}{3}$ and $-\frac{\pi}{4} \leq f(x) \leq \frac{\pi}{4}$
(D) $-\frac{1}{3} \leq x \leq \frac{1}{3}$ and $-\pi \leq f(x) \leq \pi$

7. A cubic equation has only one real root.

Which of the following statement(s) is/are true about the equation:

- I It has neither a maximum nor a minimum value
II Its maximum and minimum values must have opposite signs
III It must have a horizontal point of inflexion

- (A) I
(B) II
(C) III
(D) I or III

8. A projectile is fired from point O with an initial velocity of $V \text{ ms}^{-1}$ at an angle of θ with the horizontal. At time t seconds its position is given by

$$x = 100t \text{ and } y = 173t - 5t^2$$

The initial velocity V (correct to 3 significant figures) is:

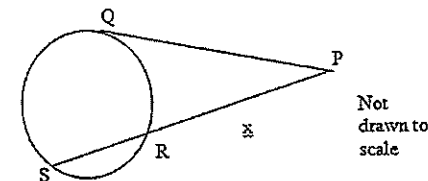
- (A) 190 ms^{-1}
(B) 200 ms^{-1}
(C) 210 ms^{-1}
(D) 215 ms^{-1}

9. PQ is a tangent to the circle where PRS is a secant intersecting the circle

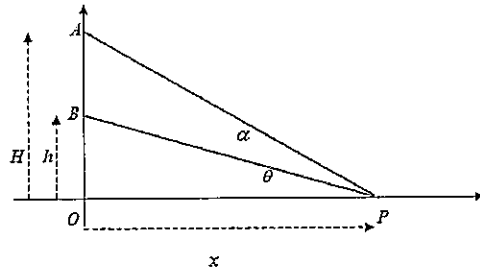
at S and R. It is given that $SR = 5$ units, $QP = 6$ units and $RP = x$ units.

The value of x is:

- (A) 1 unit
(B) $\sqrt{11}$ units
(C) 4 units
(D) 5 units



10. The diagram shows point P on the horizontal axis, a variable distance x from the origin O . The points A and B are fixed points on the vertical axis, with distance H and h respectively, from the origin O .



Let $\angle BPO = \theta$ and $\angle APB = \alpha$. Then the value of α is?

- (A) $\tan^{-1}\left(\frac{x}{H}\right) - \tan^{-1}\left(\frac{x}{h}\right)$
 (B) $\tan^{-1}\left(\frac{x}{H}\right) + \tan^{-1}\left(\frac{x}{h}\right)$
 (C) $\tan^{-1}\left(\frac{H}{x}\right) - \tan^{-1}\left(\frac{h}{x}\right)$
 (D) $\tan^{-1}\left(\frac{H}{x}\right) + \tan^{-1}\left(\frac{h}{x}\right)$

End of multiple choice section

Question 11 (15 marks) Use a SEPARATE writing booklet.

Marks

- (a) For the function $f^{-1}(x) = 3 \cos^{-1}(1-x)$
- (i) write the domain and range 2
- (ii) sketch the graph 2
- (iii) write down the equation of $f(x)$ 2

- (b) Find $\frac{d}{dx}(e^{2x} \cos^{-1} x)$ 2

- (c) (i) Express $\sin 2x - \cos 2x$ in the form $R \sin(2x - \alpha)$ 2
- (ii) Hence or otherwise, sketch the graph of $y = \sin 2x - \cos 2x$ 2

$$\text{for } -\frac{\pi}{2} \leq x \leq \frac{\pi}{2}$$

- (d) Use mathematical induction to prove 3

$$\frac{1}{2} + \frac{2}{2^2} + \frac{3}{2^3} + \dots + \frac{n}{2^n} = 2 - \frac{n+2}{2^n}$$

End of question 11

Question 12 (15 marks) Use a SEPARATE writing booklet.

Marks

- (a) Integrate $\int \frac{1}{1+16x^2} dx$ 2
- (b) A particles displacement is $x = 6 - 2\cos 2t$, where x is in centimetres and t is in seconds.
- (i) Show that the motion is simple harmonic. 2
- (ii) What are the amplitude, period and centre of the motion? 2
- (iii) Find the time and position when the acceleration is first maximum. 2
- (iv) Find first two times its speed is half its maximum speed. 2
- (c) Explain why $\sin^{-1} \sin 2 \neq 2$ 1
- (d) Find the exact value of $\sin^{-1} \left(\sin \frac{4\pi}{3} \right)$ 1
- (e) Show that $y = e^{-x} - 2x$ has only one real root. 3
The root lies in the interval $0.3 < x < 0.4$. Taking $x_1 = 0.35$ as a first approximation, use one application of Newton's method, show that a better approximation x_2 , correct to 3 significant figures, is 0.352

End of question 12

Question 13 (15 marks) Use a SEPARATE writing booklet

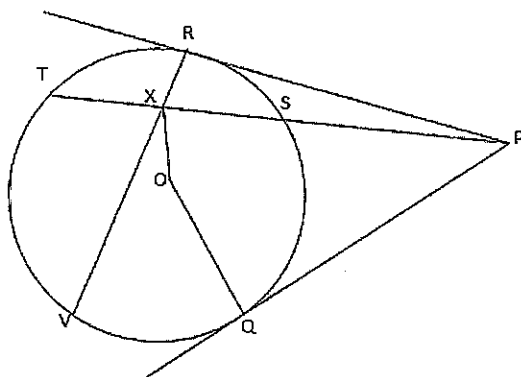
- (a) Find $\int \sin^2 5x dx$ 2
- (b) A function $f(x)$ is defined by $f(x) = x^2 - kx + 3$, where k is a non-zero constant.
- (i) Give a reason why the inverse function of $f(x)$ does not exist. 1
- (ii) Given that $f(x) \geq -1$ and that $f^{-1}(x)$ exists if the domain is restricted to the set of negative real numbers, find the value of k . 1
- (iii) Write down the equation of $f^{-1}(x)$. 1
- (c) Carbon-14 is a radioactive isotope. The amount of present day levels of it is used to find the age of the ancient remains. It's rate of disintegration is given by $\frac{dN}{dt} = -kN$, where N is the amount or concentration of Carbon-14 and k is a constant.
- (i) Show all steps in solving the differential equation above to give $N = N_0 e^{-kt}$, where N_0 is the initial amount. 2
(No marks will be given for differentiating the solution to show that it satisfies the differential equation.)
- (ii) The half-life period, denoted by $t_{1/2}$ of a radioactive element is the time taken for the amount or concentration of that isotope to fall to half of its original value. 1
If $t_{1/2}$ of Carbon-14 is 5570 years, find the exact value of k .
- (iii) Central American civilisations include the following: 2
Mayan 2000 BC – 900 AD, Toltec 900-1200AD, Aztec 1200-1500 AD. Identify the civilisation whose remains contain a Carbon-14 level of 75% of present day levels

Question 13 is continued on the next page

Question 13 continued

- (d) If the roots of the equation $32x^3 - 48x^2 + 22x + 24 = 0$, form consecutive terms of an arithmetic sequence, find one of the roots. 1

(e)



In the diagram above, O is the centre of the circle. From a point P, tangents are drawn to the circle touching the circle at Q and R. A line through P cuts the circle at S and T and OX bisects ST. RX produced cuts the circle at V.

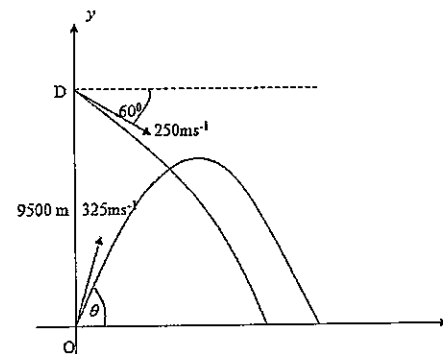
- (i) Explain why OXRP is a cyclic quadrilateral. 1
- (ii) Prove that TS is parallel to VQ. 3

End of question 13

Question 14 (15 marks) Use a SEPARATE writing booklet

Marks

- (a) During an army exercise, a surface to air missile is launched from the point O in order to intercept a dummy bomb that is released from point D. The point D is 9500 metres directly above O.



The dummy bomb is released at an angle of 60° below the horizontal with a velocity of 250 ms^{-1} . It can be shown that the equations of motion of the dummy bomb are :

$$x_D = 125t \quad \text{and} \quad y_D = 9500 - 125\sqrt{3}t - 5t^2 \quad (\text{Do NOT prove these results})$$

- (i) Calculate how long it would take the dummy bomb to reach the ground (correct to the nearest second) and where it would strike the ground (correct to the nearest minute). 2

The missile is launched at the same time as the dummy is released. It is launched with an initial velocity of 325 ms^{-1} and its angle of projection above the horizontal is θ .

The equations of motion of the missile are:

$$x_M = 325t \cos \theta \quad \text{and} \quad y_M = 325t \sin \theta - 5t^2 \quad (\text{Do NOT prove these results})$$

- (ii) Show that in order for the missile to intercept the dummy bomb it must be launched with an angle of projection of $\theta = \cos^{-1}\left(\frac{5}{13}\right)$. 1
- (iii) How high above the ground, correct to the nearest metre, does the collision occur? 3

Question 14 is continued on the next page

Question 14 continued

- (b) The acceleration (in ms^{-2}) of a particle P is given by the equation

$$\frac{d^2x}{dt^2} = 2x^3 + 4x.$$

where x is the displacement of P from a fixed point O after t seconds.

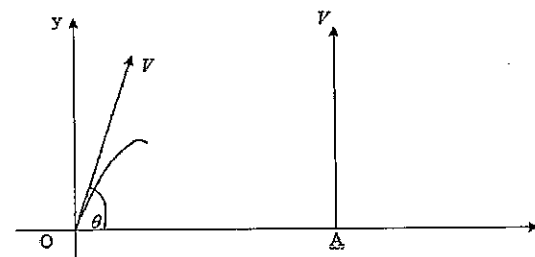
- (i) If the particle is initially 2 metres to the right of O travelling with velocity 6 ms^{-1} , find an expression v^2 (the square of the velocity) in terms of x . 2
- (ii) What is the minimum speed of the object? 2
Give a reason for your answer.

Question 14 is continued on the next page

Question 14 continued

- (c) Two particles are fired simultaneously from the ground at time $t = 0$. Particle 1 is projected from the origin at an angle of θ , $0 < \theta < \frac{\pi}{2}$, with an initial velocity V .

Particle 2 is projected vertically upward from point A, a distance α to the right of the origin, also with an initial velocity V .



It can be shown that while both particles are in flight, Particle 1 has

equations of motion: $x = Vt \cos \theta$ $y = Vt \sin \theta - \frac{1}{2}gt^2$

and Particle 2 has equations of motion: $x = \alpha$ $y = Vt - \frac{1}{2}gt^2$

(Do NOT prove these equations of motion).

Let L be the distance between the particles at time t .

- (i) Show that while both particles are in flight, 2
 $L^2 = 2V^2t^2(1 - \sin \theta) - 2\alpha Vt \cos \theta + \alpha^2.$
- (ii) An observer notices that the distance between the particles in flight first decreases then increases. 3

Show that the distance between the particles in flight is smallest when

$$t = \frac{\alpha \cos \theta}{2V(1 - \sin \theta)} \text{ and that the smallest distance is } \alpha \sqrt{\frac{1 - \sin \theta}{2}}.$$

End of Paper



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Multiple Choice Answer Sheet

Completely fill the response oval representing the most correct answer.

1. A ☐ B ☐ C ☐ D ☐

2. A ☐ B ☐ C ☐ D ☐

3. A ☐ B ☐ C ☐ D ☐

4. A ☐ B ☐ C ☐ D ☐

5. A ☐ B ☐ C ☐ D ☐

6. A ☐ B ☐ C ☐ D ☐

7. A ☐ B ☐ C ☐ D ☐

8. A ☐ B ☐ C ☐ D ☐

9. A ☐ B ☐ C ☐ D ☐

10. A ☐ B ☐ C ☐ D ☐

STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right)$$

NOTE: $\ln x = \log_e x, \quad x > 0$

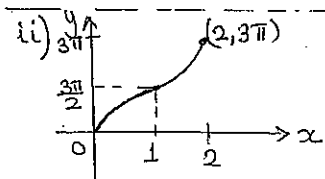
Apple Choice:

1. B, 2D, 3C, 4B, 5A, 6D, 7D, 8B, 9C, 10C

SOLUTIONS

ii) a) $f^{-1}(x) = 3\cos^{-1}(1-x)$

i) Domain $-1 \leq 1-x \leq 1$, Range $0 \leq \cos^{-1}(1-x) \leq \pi$
 $-2 \leq -x \leq 0$ $0 \leq 3\cos^{-1}(1-x) \leq 3\pi$
 $0 \leq x \leq 2$ or $0 \leq y \leq 3\pi$



iii) $f^{-1}(x) = 3\cos^{-1}(1-x)$
 $x = 3\cos^{-1}(1-y)$
 $\frac{x}{3} = \cos^{-1}(1-y)$
 $\cos \frac{x}{3} = 1-y$
 $y = 1 - \cos \frac{x}{3}$

b) $\frac{d}{dx}(e^{2x} \cos^{-1} x) = 2e^{2x} \cos^{-1} x - \frac{e^{2x}}{\sqrt{1-x^2}}$

c) i) $\sin 2x - \cos 2x = R \sin(2x - \alpha)$

where $a=1, b=1$ OR $\sin 2x - \cos 2x = R \sin(2x - \alpha)$
 $R = \sqrt{a^2 + b^2} = \sqrt{2}$ $\Rightarrow R \cos \alpha = 1, R \sin \alpha = 1$
 $\tan \alpha = \frac{b}{a} = 1$ square and add
 $R^2 \cos^2 \alpha = 1, R^2 \sin^2 \alpha = 1$
 $R^2 (\sin^2 \alpha + \cos^2 \alpha) = 2$
 $R = \sqrt{2}$
 $\frac{1}{\sqrt{2}} \tan \alpha = 1, \alpha = \frac{\pi}{4}$

$\therefore \sin 2x - \cos 2x = \sqrt{2} \sin(2x - \frac{\pi}{4})$

Award

2 marks correct domain and range.

1 mark correct domain

1 mark correct range

2 marks correct shape correct boundaries.

1 mark some working towards solution

2 marks correct equation of $f(x)$

1 mark $\frac{x}{3} = \cos^{-1}(1-y)$

2 marks correct answer
1 mark correct differentiation of e^{2x} or $\cos^{-1} x$

2 marks correct R, α and expression

1 mark correct R or correct α

$\frac{1}{\sqrt{2}} \sin 2x - \frac{1}{\sqrt{2}} \cos 2x$
 $\sqrt{2} \left[\frac{1}{\sqrt{2}} \sin 2x - \frac{1}{\sqrt{2}} \cos 2x \right]$
 $\sqrt{2} \left[\cos \frac{\pi}{4} \sin 2x - \sin \frac{\pi}{4} \cos 2x \right]$
 $\sqrt{2} \left[\sin(2x - \frac{\pi}{4}) \right]$

Q11 c ii continue

Hence the statement is true for $n=k$ then it is true for $n=k+1$.

Thus since S(1) is true then S(2) must be true.

If S(2) is true, then S(3) must be true and so on.

Q12 a) $\int \frac{1}{1+16x^2} dx = \frac{1}{16} \int \frac{1}{\frac{1}{16} + x^2} dx$ where $a^2 = \frac{1}{16}$
 and $a = \frac{1}{4}$

$= \frac{1}{16} \int \frac{\frac{1}{4}}{\frac{1}{4}(\frac{1}{16} + x^2)} dx$
 $= \frac{1}{4} \tan^{-1} 4x + C$

b) $x = 6 - 2\cos 2t \Rightarrow 2\cos 2t = 6 - x$

i) $\dot{x} = 4\sin 2t$ velocity

$\ddot{x} = 8\cos 2t$

$\dot{x} = 4(6-x)$

$\ddot{x} = -4(x-6)$

If $x-6=y$ then $\dot{x}=\dot{y}$ and $\ddot{x}=\ddot{y}$ (*important)

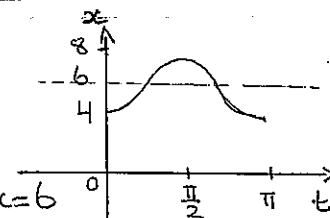
then $\ddot{y} = -4y$ This is $\ddot{y} = -n^2 y$ format

therefore motion is simple harmonic where

$n^2 = 4$ and $n = 2, n > 0$

ii) Amplitude = 2,
Period $\frac{2\pi}{2} = \pi$

Centre of the motion $x = 6$



2 marks correct integration

1 mark any attempt towards solution

Do not penalise for not writing 'C'.

2 marks

Establishes that

$\ddot{x} = -4x$ or

$\ddot{y} = -4y$

1 mark $\ddot{x} = -4(x-6)$

2 marks for 3 properties

1 mark 2 properties established.

Q12 b continues

iii) $\ddot{x} = 8 \cos 2t$

$-1 \leq \cos 2t \leq 1$

$-8 \leq 8 \cos 2t \leq 8$

Acceleration is maximum when $\cos 2t = 1$

$\therefore$ Maximum $\ddot{x} = 8 \text{ cm/sec}^2$

$8 = 8 \cos 2t$ when $t = t \text{ sec}$

$\cos 2t = 1$

$2t = 2\pi$

$t = \pi \text{ sec}$

$x = 6 - 2 \cos 2\pi$

$x = 4 \text{ cm}$

iv) Maximum speed when $\sin 2t = 1$, as $-1 \leq \sin 2t \leq 1$

$\therefore \dot{x} = 4 \sin 2t = 4$, then half maximum speed = 2 cm/s

$2 = 4 \sin 2t$

$\sin 2t = \frac{1}{2}$

$2t = \frac{\pi}{6}, \frac{5\pi}{6}$

$t = \frac{\pi}{12}, \frac{5\pi}{12} \text{ s}$

c) $\sin^{-1}(\sin 2) \neq 2$, $y = \sin^{-1}x$ is restricted $-\frac{\pi}{2} \leq y \leq \frac{\pi}{2}$

$2 > \frac{\pi}{2}$ therefore $\sin^{-1} \sin 2 \neq 2$

d) $\sin^{-1}(\sin \frac{4\pi}{3}) = \sin^{-1}(-\frac{\sqrt{3}}{2}) = -\frac{\pi}{3}$

e) $y = e^{-x} - 2x$, to show 'y' has only one root, show that it does not have turning points, i.e. it is monotonic increasing or decreasing.

$y' = -e^{-x} - 2$ at turning point $y' = 0$

$e^{-x} = -2$, $\frac{1}{e^x} = -2$ not true as $\frac{1}{e^x} > 0$

To find an approximation:

$x_2 = x_1 - \frac{f(x_1)}{f'(x_1)}$

$x_1 = 0.35$ $f(0.35) = \frac{1}{e^{0.35}} - 0.7$

$f'(0.35) = -\frac{1}{e^{0.35}} - 2$

$x_2 = 0.35 - \frac{\frac{1}{e^{0.35}} - 0.7}{-\frac{1}{e^{0.35}} - 2}$

$x_2 = 0.351733...$

$x_2 = 0.352$

forward

1 mark establishes that $\ddot{x} = 8 \text{ cm/sec}^2$

1 mark for finding time and position

1 mark establishes that half of maximum speed is 2 cm/s .

1 mark finds $t = \frac{\pi}{12}, \frac{5\pi}{12} \text{ s}$

1 mark correct explanation

1 mark must show work

1 mark shows that $y' = -e^{-x} - 2$

and $\frac{1}{e^x}$ is always positive. $\therefore \frac{1}{e^x} \neq -2$

1 mark substituting into formula correctly

1 mark correct answer (3 sig. fig)

Question 13

(a) $\int \sin^2 5x \, dx$
 $= \int \frac{1}{2} (1 - \cos 10x) \, dx$ ← correct conversion to double angle 1 mark
 $= \frac{1}{2} (x - \frac{1}{10} \sin 10x) + C$ ← correct answer 1 mark

(b) i) $f(x) = x^2 - kx + 3$
 $f(x) = 3 \Rightarrow x = 0$ or $x = k$
 $\therefore$ for a given value of $f(x)$, there are two distinct values of x ($k \neq 0$)
 $\therefore f(x)$ doesn't have an inverse 1 mark

or any other suitable explanation
 $f(x) \geq -1 \Rightarrow \frac{12 - k^2}{4} \geq -1$ 1 mark

$f(x) \geq -1$
 $\therefore y = -1$ is turning pt. (ii)
 $x = -\frac{b}{2a} = -\frac{-k}{2} = \frac{k}{2}$

sub into $f(x)$
 $-1 = \frac{k^2}{4} - \frac{k^2}{2} + 3$ $\frac{-k^2}{4} + 4 = 0$

$k^2 = 16$
 $k = \pm 4$

domain is restricted to negative real no's.

$k = -4$ $x = \frac{-4}{2} = -2 \therefore x \leq -2$

$\Rightarrow k = \pm 4$
 min. value of $f(x)$ occurs at
 $x = \frac{k}{2} < 0$
 $\therefore k = -4 \Rightarrow x \leq -2$

when $k = -4$
 $f(x) = x^2 + 4x + 3$
 $x = [f^{-1}(y)]^2 + 4f^{-1}(y) + 3$ 1 mark
 $\Rightarrow f^{-1}(x) = -2 \pm \sqrt{1+x}$
 since $f^{-1}(x) \leq -2$
 $f^{-1}(x) = -2 - \sqrt{1+x}$

(c) (i) $\frac{dN}{dt} = -kN$

$\int \frac{dN}{N} = \int -k dt$ — 1 mark

$\ln N = -kt + C$ — 1 mark

$t=0, N=N_0$

$\therefore C = \ln N_0$

$\ln\left(\frac{N}{N_0}\right) = -kt$

$N = N_0 e^{-kt}$

(ii) $N = N_0 e^{-kt}$

$\frac{N_0}{2} = N_0 e^{-kt}$

$\Rightarrow e^{kt} = 2$

$\Rightarrow t_{\frac{1}{2}} = \frac{1}{k} \ln 2 = 5570$ — correct answer 1 mark

~~$k = \frac{\ln 2}{5570}$~~

(iii) $75 = 100 e^{-kt}$

$t = \frac{1}{k} \ln\left(\frac{4}{3}\right)$

$= \frac{5570}{\ln 2} \ln\left(\frac{4}{3}\right)$

$= 2311.76 \sim 2312$ yrs. Mayan civilisation

(d) $32x^3 - 48x^2 + 22x + 24 = 0$

let the roots be $a-d, a, a+d$

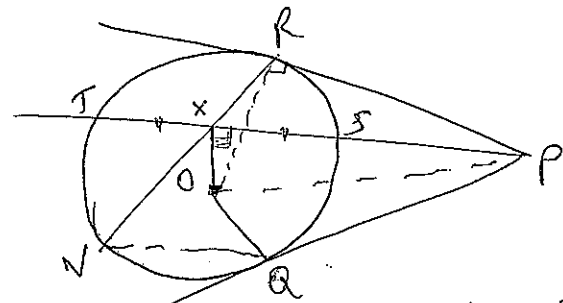
sum of roots $= -\left(\frac{-48}{32}\right)$

$\Rightarrow 3a = \frac{48}{32}$

$\Rightarrow a = \frac{1}{2}$

— 1 mark for correct marking and correct answer

(e)



(i) $\angle OXP = 90^\circ$ (angle between a chord and the line joining the mid point of the chord to the centre of the circle) — 1 mark
(or equivalent statement)

$\angle ORP = 90^\circ$ (PR is a tangent) (O is the centre of the circle) $\therefore OR \perp PR$

Since $\angle ORP = \angle OXP = 90^\circ$
OXRP is a cyclic quadrilateral with OP as a diagonal (angles subtended by angles in semi-circle of an eqn) — 1 mark

(ii) let $\angle PXR = \theta$
 $\therefore \angle POR = \theta$ ($\because$ OXRP is a cyclic quad)
But $\angle POR = \angle POQ$ ($\because$ PR and PQ are tangents)

$\Rightarrow \angle ROQ = 2\theta$

$\Rightarrow \angle RQA = \theta$ (angle subtended by arc RQA at the centre is double the angle subtended by the same arc on the circumference)

OR $\angle PXR = \theta$

$\angle RQA = \theta$

$\angle PXR = \angle RQA = \theta$

Corresponding $\angle$'s $\therefore TS \parallel VQ$

$\angle TXV = \angle PXR = \theta$ (vert. opp. $\angle$'s)

$\angle TXV = \angle RQA = \theta$

$\Rightarrow TS \parallel VQ$ (alt. angles $\angle TXV = \angle RQA$ are eqn)

Question 14

(a) (i) Dummy bomb reaches ground when $y_0 = 0$

$$\text{i.e. } 0 = 9500 - 125\sqrt{3}t - 5t^2$$

$$t^2 + 25\sqrt{3}t - 1900 = 0$$

$$t = \frac{-25\sqrt{3} \pm \sqrt{625 \times 3 + 7600}}{2}$$

$$= \frac{-25\sqrt{3} + \sqrt{7625}}{2} \quad (\text{ignore } (-t))$$

$$\div 27.61917074$$

$$t \div 27 \text{ seconds}$$

$$\text{when } t = 27$$

$$x_0 = 125 \times 27$$

$$= 3375 \text{ metres from } 0.$$

(ii) for intersection to occur $x_b = x_m$

$$\text{i.e. } 125t = 325t \cos \theta$$

$$125 = 325 \cos \theta$$

$$\cos \theta = \frac{125}{325} = \frac{5}{13}$$

$$\therefore \theta = \cos^{-1} \frac{5}{13}$$

Question 14

(a) (iii)

Collision occurs when $y_b = y_m$

$$\text{i.e. } 9500 - 125\sqrt{3}t - 5t^2 = 325t \sin \theta - 5t^2$$

$$325t \sin \theta + 125\sqrt{3}t = 9500$$

$$t(325 \sin \theta + 125\sqrt{3}) = 9500$$

$$t = \frac{9500}{325 \sin \theta + 125\sqrt{3}}$$

$$= \frac{9500}{325 \sin(\cos^{-1} \frac{5}{13}) + 125\sqrt{3}}$$

$$= \frac{9500}{325 \times \frac{12}{13} + 125\sqrt{3}}$$

$$= \frac{9500}{325 \times \frac{12}{13} + 125\sqrt{3}}$$

$$t \div 18.4 \text{ sec}$$

$$\text{Height} = 325 \times t \times \sin \theta - 5t^2$$

$$= 325 \times (18.4) \times \frac{12}{13} - 5(18.4)^2$$

$$= 3827 \text{ m}$$

1 for showing $y_b = y_m$ (or the equation).

1 mark for $t = 18.4$. (or equivalent, may be exact value).

1 mark for 3827m (if exact + used it is 3826m).

Question 14

(b) (i) $\frac{d^2x}{dt^2} = 2x^3 + 4x$; $t=0$, $x=2$, $v=6$

$$\frac{d(\frac{1}{2}v^2)}{dx} = 2x^3 + 4x$$

$$d(\frac{1}{2}v^2) = (2x^3 + 4x) dx$$

$$\int d(\frac{1}{2}v^2) = \int (2x^3 + 4x) dx$$

$$\frac{1}{2}v^2 = \frac{1}{2}x^4 + 2x^2 + C$$

$$\frac{1}{2} \times (6)^2 = \frac{1}{2} \times (2)^4 + 2 \times (2)^2 + C; x=2, v=6$$

$$18 = 8 + 8 + C$$

$$C = 2$$

$$\therefore \frac{1}{2}v^2 = \frac{1}{2}x^4 + 2x^2 + 2$$

$$v^2 = x^4 + 4x^2 + 4$$

(ii) $v^2 = x^4 + 4x^2 + 4$

$$v^2 = (x^2 + 2)^2$$

The minimum value for v^2 occurs where $x^2 = 0$

This equates to $v^2 = 4$

$$\text{ie } v = \pm 2$$

$\therefore$ the minimum speed is 2 ms^{-1} .

1 for knowing
to write
 $\frac{d^2x}{dt^2}$ as $\frac{d(\frac{1}{2}v^2)}{dx}$

$$\text{ie } \frac{d(\frac{1}{2}v^2)}{dx} = 2x^3 + 4x$$

1 for answer

1 for explaining
why $v^2 = 4$

1 for answer

Question 14

(c) (i)

Particle 1 : $A(vt \cos \theta, vt \sin \theta - \frac{gt^2}{2})$

Particle 2 : $B(\alpha, vt - \frac{gt^2}{2})$

Let L be the distance between A and B

$$L = \sqrt{(vt \cos \theta - \alpha)^2 + (vt \sin \theta - \frac{gt^2}{2} - vt + \frac{gt^2}{2})^2}$$

$$= \sqrt{v^2 t^2 \cos^2 \theta + \alpha^2 - 2\alpha vt \cos \theta + (vt \sin \theta - vt)^2}$$

$$= \sqrt{v^2 t^2 \cos^2 \theta + \alpha^2 - 2\alpha vt \cos \theta + v^2 t^2 \sin^2 \theta - 2v^2 t \sin \theta + v^2 t^2}$$

$$= \sqrt{2v^2 t^2 - 2\alpha vt \cos \theta - 2v^2 t^2 \sin \theta + \alpha^2}$$

$$L^2 = 2v^2 t^2 (1 - \sin \theta) - 2\alpha vt \cos \theta + \alpha^2$$

(ii) L^2 is a quadratic in terms of t (concave up)

$\therefore$ Min value occurs at $t = -\frac{b}{2a}$

$$\text{ie } t = \frac{2\alpha v \cos \theta}{2(2v^2)(1 - \sin \theta)} = \frac{\alpha \cos \theta}{2v(1 - \sin \theta)}$$

sub into L^2 gives

$$L^2 = 2v^2 \left(\frac{\alpha^2 \cos^2 \theta}{4v^2 (1 - \sin \theta)^2} \right) (1 - \sin \theta) - 2\alpha v \left(\frac{\alpha \cos \theta}{2v(1 - \sin \theta)} \right) \frac{\cos \theta}{1} + \alpha^2$$

$$= \frac{\alpha^2 \cos^2 \theta}{2(1 - \sin \theta)} - \frac{2\alpha^2 \cos^2 \theta}{2(1 - \sin \theta)} + \alpha^2$$

$$= \frac{-\alpha^2 \cos^2 \theta}{2(1 - \sin \theta)} + \alpha^2$$

$$= \alpha^2 \left(1 - \frac{\cos^2 \theta}{2(1 - \sin \theta)} \right)$$

1 for this line
1 for working
out to arrive
at final answer.

1 for explaining
and arriving
at t value.

1 for sub into
 L^2 (this line)

1 for working
to arrive at
final answer.

Question 14.

(c) (ii) continued

$$L = \alpha \sqrt{1 - \frac{(1 - \sin \theta)(1 + \sin \theta)}{2(1 - \sin \theta)}}$$

$$= \alpha \sqrt{1 - \frac{1 + \sin \theta}{2}}$$

$$= \alpha \sqrt{\frac{1 - \sin \theta}{2}}$$

(or)

We can find $\frac{d(L^2)}{dt}$ and show that a stationary point exists at

$$t = \frac{\alpha \cos \theta}{2V(1 - \sin \theta)} \quad \text{then substitute into}$$

$$L^2.$$

End.